

SM015
Mathematics 1
Semester I
Session 2021/2022
1½ hours



KEMENTERIAN PENDIDIKAN MALAYSIA

BAHAGIAN MATRIKULASI

CUMULATIVE TEST MATRICULATION PROGRAMME

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INSTRUCTIONS TO CANDIDATE:

This question paper consists of **Section A** and **Section B**.

Section A consists of **10** questions. **Section B** consists of **20** questions.

Answer **all** questions.

The use of non-programmable scientific calculator is permitted.

LIST OF MATHEMATICAL FORMULAE
SENARAI RUMUS MATEMATIK

Indices*Indeks*

$$a^m \times a^n = a^{m+n}$$

$$a^m \div a^n = a^{m-n}$$

$$(ab)^m = a^m b^m$$

Logarithms*Logaritma*

$$\log ab = \log a + \log b$$

$$\log \left(\frac{a}{b} \right) = \log a - \log b$$

$$\log a^b = b \log a$$

Surds*Surds*

$$\sqrt{a} \times \sqrt{b} = \sqrt{ab}$$

$$\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}$$

Quadratic equation $ax^2 + bx + c = 0$:

Persamaan kuadratik $ax^2 + bx + c = 0$:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Absolute Values*Nilai Mutlak*

$$|x| = a \Leftrightarrow x = a \text{ or } x = -a$$

$$|x| < a \Leftrightarrow -a < x < a$$

$$|x| > a \Leftrightarrow x < -a \text{ or } x > a$$

LIST OF MATHEMATICAL FORMULAE
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Trigonometry
Trigonometri

$$\sin (A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos (A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan (A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A$$

$$= 2 \cos^2 A - 1$$

$$= 1 - 2 \sin^2 A$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$$\sin^2 A = \frac{1 - \cos 2A}{2}$$

$$\cos^2 A = \frac{1 + \cos 2A}{2}$$

LIST OF MATHEMATICAL FORMULAE
SENARAI RUMUS MATEMATIK

Differentiation
Pembezaan

$f(x)$	$f'(x)$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\cot x$	$-\operatorname{cosec}^2 x$
$\sec x$	$\sec x \tan x$
$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$

The Sum and Difference Rule

If $f(x) = u(x) \pm v(x)$,
 then $f'(x) = u'(x) \pm v'(x)$

Product Rule and Quotient Rule

$$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

The Chain Rule

If $y = f(u)$ and $u = g(x)$,

then $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

If $y = \ln f(x)$,

then $\frac{dy}{dx} = \frac{1}{f(x)} \cdot f'(x)$

LIST OF MATHEMATICAL FORMULAE
SENARAI RUMUS MATEMATIK

If $y = a^{f(x)}$

then $\frac{dy}{dx} = a^{f(x)} f'(x) \ln a$

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$$

$$\frac{d^2 y}{dx^2} = \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{\frac{dx}{dt}} \quad \text{OR} \quad \frac{d^2 y}{dx^2} = \left[\frac{d}{dt} \left(\frac{dy}{dx} \right) \right] \times \frac{dt}{dx}$$

Limit and Continuity
Had dan Keselajaran

Vertical Asymptotes

If $\lim_{x \rightarrow a^+} f(x) = \pm\infty$ and/or $\lim_{x \rightarrow a^-} f(x) = \pm\infty$, $x = a$ is a vertical asymptote of $f(x)$.

Horizontal Asymptotes

If $\lim_{x \rightarrow \infty} f(x) = L$ and/or $\lim_{x \rightarrow -\infty} f(x) = L$, then L is a horizontal asymptote of $f(x)$.

Sphere
Sfera

$$V = \frac{4}{3} \pi r^3$$

$$S = 4 \pi r^2$$

Right circular cone
Kon membulat tegak

$$V = \frac{1}{3} \pi r^2 h$$

$$S = \pi r^2 + \pi r h$$

Right circular cylinder
Silinder membulat tegak

$$V = \pi r^2 h$$

$$S = 2\pi r^2 + 2\pi r h$$

SECTION A

This section consists of **10** questions. Each question has **four** answer options. Answer **all** questions. Choose the **best** answer. Shade your answers on the given answer sheet.

1 Find $\lim_{x \rightarrow 3} \frac{x^2 - 8x + 15}{x - 3}$.

- A -2
- B 0
- C -3
- D Undefined

2 Function f is discontinuous at $x = a$. Identify the possible reasons of the discontinuity.

- I $f(a)$ is not defined.
- II $\lim_{x \rightarrow a} f(x)$ does not exist.
- III $\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x)$.
- IV $f(a) \neq \lim_{x \rightarrow a} f(x)$.

- A II and III
- B I and IV
- C I, II and IV
- D II, III and IV

3 Which of the following will indicate the existence of limit?

A $\lim_{x \rightarrow 0^+} \frac{x-1}{x^2}$

B $\lim_{x \rightarrow 1^-} \sqrt{x-1}$

C $\lim_{x \rightarrow 2} \frac{4-x^2}{x-2}$

D $\lim_{x \rightarrow 0} \frac{|x|}{x}$

4 Determine the **CORRECT** method to find the derivative of $f(x) = \sqrt{x-1}$ by using the first principles.

A $\lim_{x \rightarrow 0} \frac{\sqrt{x-1+h} - \sqrt{x-1}}{h}$

B $\lim_{h \rightarrow 0} \frac{\sqrt{x-1+h} - \sqrt{x-1}}{h}$

C $\lim_{x \rightarrow 0} \frac{\sqrt{x-1-h} - \sqrt{x-1}}{h}$

D $\lim_{h \rightarrow 0} \frac{\sqrt{x-1-h} - \sqrt{x-1}}{h}$

- 5 Given that $y = \frac{P}{x^2} + \frac{3Q}{x}$, where P and Q are constants and $x \neq 0$. Find $\frac{d^2y}{dx^2}$.

A $\frac{d^2y}{dx^2} = \frac{6}{x^4}(P + Qx)$

B $\frac{d^2y}{dx^2} = \frac{2}{x^2}\left(P + \frac{3Q}{x}\right)$

C $\frac{d^2y}{dx^2} = \frac{6}{x^4}\left(3Q - \frac{4P}{x}\right)$

D $\frac{d^2y}{dx^2} = \frac{6}{x^3}\left(Q - \frac{P}{x}\right)$

- 6 Find $f'''(x)$ if $f''(x) = 4e^{1-2x}$.

A $8e^{1-2x}$

B $-4e^{1-2x}$

C $4e^{1-2x}$

D $-8e^{1-2x}$

7 Find $\frac{d}{dx}[\ln(2+x^2)]$.

A $\frac{2x}{2+x^2}$

B $\frac{2}{2+x^2}$

C $\frac{2}{x^2}$

D $\frac{1}{2+x^2}$

8 Find the gradient of the curve $y = 6 \tan \frac{1}{3}x$.

A $6 \sec^2 \frac{1}{3}x$

B $2 \sec^2 \frac{1}{3}x$

C $6 \sec^2 3x$

D $3 \sec^2 \frac{1}{3}x$

9 Find the critical numbers of the curve $f(x) = x^4 - 2x^2$.

A $x = -1, x = 1$

B $x = -1, x = 0, x = 1$

C $x = 0, x = 2$

D $x = -1, x = 1, x = 2$

- 10 The derivative of function f is given by $f'(x) = 3(x+1)(x-5)$.

Determine the x -coordinate if the stationary point exists for the function f and state its nature.

- I $x = 5$, local maximum
 - II $x = -1$, local maximum
 - III $x = 5$, local minimum
 - IV $x = -1$, local minimum
-
- A I and III
 - B II and IV
 - C I and IV
 - D II and III

SECTION B

This section consists of **20** questions. Each question has **four** answer options. Answer **all** questions. Choose the **best** answer. Shade your answers on the given answer sheet.

11 $([2, 9) \cap (-4, 7)) \cup \{0, 4, 10\}$ can be simplified as

A $[2, 7).$

B $[2, 7] \cup \{0, 4, 10\}.$

C $[2, 7) \cup \{0, 10\}.$

D $[2, 9].$

12 Find the values of a and b that satisfy the equation $2 + ai = \frac{6 - 2i}{b + i}.$

I $a = -2, b = 2$

II $a = -4, b = 1$

III $a = 0, b = 1$

IV $a = -1, b = 3$

A I and II

B I, III and IV

C IV only

D All of the above

13 Solve the inequality $5x > 2x^2 + 3$.

A $(-\infty, \infty)$

B $\left[1, \frac{3}{2}\right]$

C $\left(1, \frac{3}{2}\right)$

D $(-\infty, 1) \cup \left(\frac{3}{2}, \infty\right)$

14 Given $\frac{2x-1}{x-1} - 3 \geq 0$ and the solution is $(1, 2]$. Given $\frac{2x-1}{x-1} + 3 \leq 0$ and the solution is $\left[\frac{4}{5}, 1\right)$. Find $\left|\frac{2x-1}{x-1}\right| \geq 3$.

A $\left[\frac{4}{5}, 1\right) \cup (1, 2]$

B $\left(\frac{4}{5}, 1\right) \cup (1, 2)$

C $(-\infty, \infty)$

D $\left(-\infty, \frac{4}{5}\right) \cup (1, 2)$

15 Solve the equation $2(3^{2x+1}) + 7(3)^x - 3 = 0$.

A $x = -1$

B $x = -1, x = -\frac{3}{2}$

C $x = 1, x = -\frac{3}{2}$

D $x = 1$

16 Find the domain of $f(x) = \frac{4x-1}{\sqrt{4-x}}$.

A $(-\infty, -4) \cup (4, \infty)$

B $(-\infty, 4)$

C $(-2, 2)$

D $[-2, 2]$

17 Find $g(x)$ if $(g \circ f)(x) = \frac{1+2x}{2+x}, x \neq -2$ and $f(x) = 3+2x$.

A $g(x) = \frac{2(x+2)}{x+1}$

B $g(x) = \frac{2(x-2)}{x+1}$

C $g(x) = \frac{2(x+2)}{x-1}$

D $g(x) = \frac{2(x-2)}{x-1}$

18 Given $h(x) = e^{2x} + 3$, $x \in \mathbb{R}$. Find $h^{-1}(5)$.

A $\frac{\ln 4}{2}$

B $-\ln 2$

C $\ln 2$

D $\frac{\ln 2}{2}$

19 If $f(x) = 2x^2 + 4x - 6$, determine which of the following are **TRUE**.

I The graph's minimum point is at $(-1, -8)$.

II The intercepts of the function occur at $x = -3$ and $x = 1$.

III The domain is $(-\infty, -1)$.

IV The range is $(-8, \infty)$.

A I and II

B I and III

C I, II, and III

D All of the above

20 Determine which of the following are **TRUE** about the graph of $f(x) = -\log_{\frac{1}{5}} x$.

- I The domain is $(0, \infty)$.
- II The range is $(-\infty, \infty)$.
- III The equation of the asymptote is $x = 0$.
- IV $f(x) = -\log_{\frac{1}{5}} x = \log_5 x$.

- A I and II
- B I, II and III
- C None are true
- D All of the above

- 21 Given $f(x) = \begin{cases} \frac{x}{|x+1|}, & x \neq -1 \\ 2, & x = -1 \end{cases}$ is discontinuous at $x = -1$. Which of the following

statements are **TRUE**?

- I $f(-1) = 2$.
- II $f(-1)$ is undefined.
- III $\lim_{x \rightarrow -1} f(x)$ does not exist.
- IV $\lim_{x \rightarrow -1} f(x)$ exist.

- A I and IV
- B I and III
- C II and III
- D II and IV

- 22 Determine the vertical asymptote (if any) for the function $f(x) = \frac{3x^2 - 1}{5x^2 + 3}$.

- A $x = -\frac{3}{5}$
- B $x = \frac{3}{5}$
- C $x = 0$
- D No vertical asymptote

23 Find the horizontal asymptote(s) (if any) of $f(x) = \frac{\sqrt{4x^2 + 1}}{2x + 3}$.

- I $y = 1$
- II $y = -1$
- III $y = 0$
- IV No horizontal asymptote

- A I and II
- B II and III
- C I, II and III
- D IV only

24 Which is the **CORRECT** formula to find the value of $f'(3)$ for $f(x) = \frac{1}{\sqrt{2x+3}}$?

- A $\lim_{x \rightarrow 3} \frac{1}{3} \left(\frac{\sqrt{2x+3} - 3}{x-3} \right)$
- B $\lim_{x \rightarrow 3} \left(\frac{3 - \sqrt{2x+3}}{x-3} \right)$
- C $\lim_{x \rightarrow 3} \frac{1}{x-3} \left(\frac{3 - \sqrt{2x+3}}{\sqrt{2x+3}} \right)$
- D $\lim_{x \rightarrow 3} \frac{1}{x-3} \left(\frac{3 - \sqrt{2x+3}}{3\sqrt{2x+3}} \right)$

25 Find the derivative of $y = \ln \sqrt{2x+3}$ with respect to x in terms of y .

A $\frac{dy}{dx} = 2e^y$

B $\frac{dy}{dx} = 2e^{-y}$

C $\frac{dy}{dx} = e^{-2y}$

D $\frac{dy}{dx} = e^{2y}$

26 Given that $x = \frac{1}{1-t^2}$ and $y = \frac{1+t^2}{t}$, where t is a non-zero parameter. Which of the following are the value(s) of $\frac{dy}{dx}$ at the point $\left(-\frac{1}{3}, \frac{5}{2}\right)$?

I $-\frac{27}{16}$

II $\frac{1}{3}$

III $\frac{27}{16}$

IV $-\frac{1}{3}$

A I and III

B II and III

C II and IV

D III only

27 Given $y = 3^{2x^2+x}$, find $\frac{dy}{dx}$.

A $\ln 3(3^{2x^2+x})$

B $\ln 3(3^{2x^2+x})(4x+1)$

C $(3^{2x^2+x})(4x+1)$

D $\ln 3(4x+1)$

28 Given $6x^2 + (6y^2 - 9x)\frac{dy}{dx} - 9y = 0$ is a first derivative of a function, find $\frac{d^2y}{dx^2}$ if

$\frac{dy}{dx} = \frac{4}{5}$ is at point $(1, 2)$.

A $-\frac{324}{25}$

B $-\frac{108}{125}$

C $\frac{108}{125}$

D $\frac{324}{25}$

29 Given the equation $y = x^3 + 1$, determine the nature of its critical point (if exists).

A Relative maximum point

B Relative minimum point

C No extremum

D Does not exist

- 30 Zul is inflating a spherical balloon and the rate of change of its volume is 5 cm^3 per second. The balloon will explode if its volume reach 45 cm^3 . Assuming that initially the volume of the balloon is 0 cm^3 . Find the time when the balloon will explode after the inflation.
- A 8 seconds
 - B 9 seconds
 - C 10 seconds
 - D 11 seconds

END OF QUESTION PAPER